



Osmosis — The Story of the Shrinking Potato!





Kabir's Kitchen Poll!

Kabir put cucumber slices in salt water. What do you think will happen to them?

Will the cucumber slices become SOFT or stay FIRM?
Pick your answer!

POLL



Droopy



Cucumber



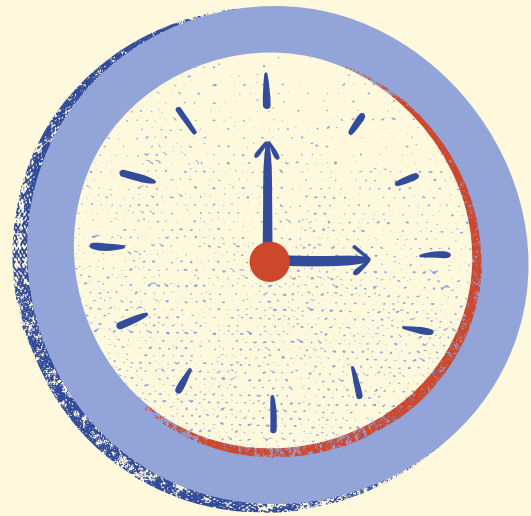
Potato Experiment!

**Cut two potato strips.
Soak one in plain water, one in salt water
(2 tbsp salt).
Watch what happens!**

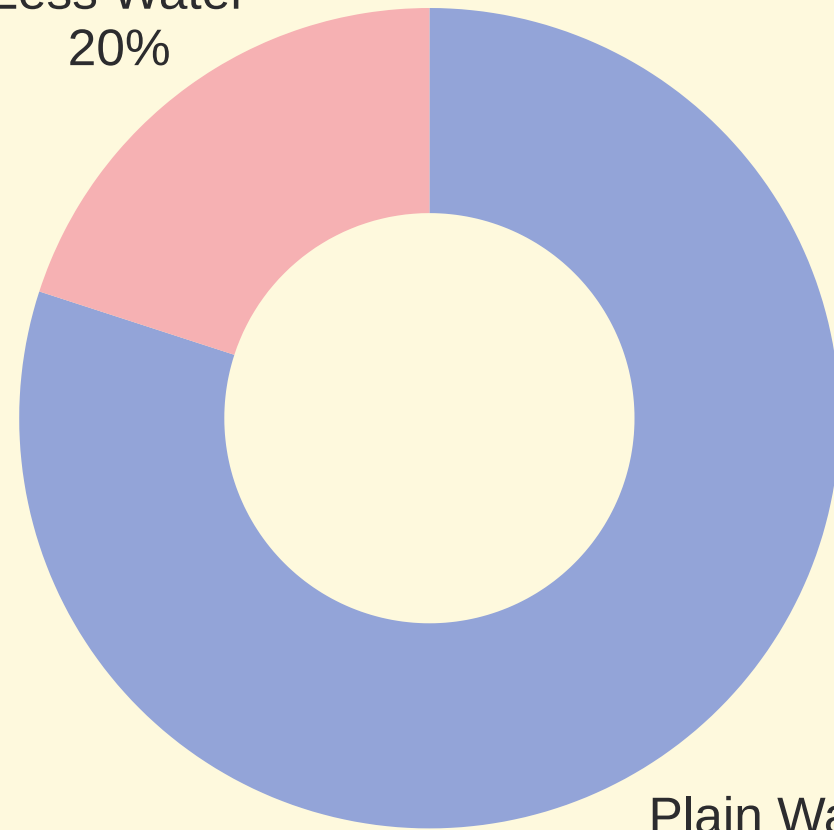
Which potato strip feels softer after
30 minutes? Why do you think so?

Plain Water vs Salt Water — Let's find out what happens to a potato strip in each! 🥔





Salt Water —
Less Water
20%



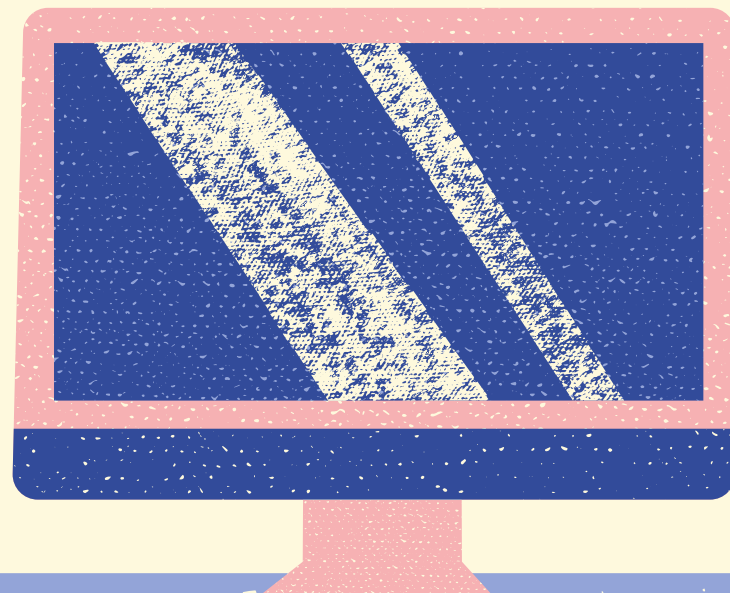
Plain Water
— More
Water
80%

Source: AI-generated data. Replace or verify
before use.

What Happens Inside? Water Moves!

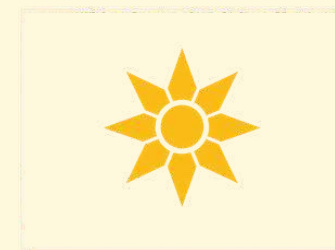
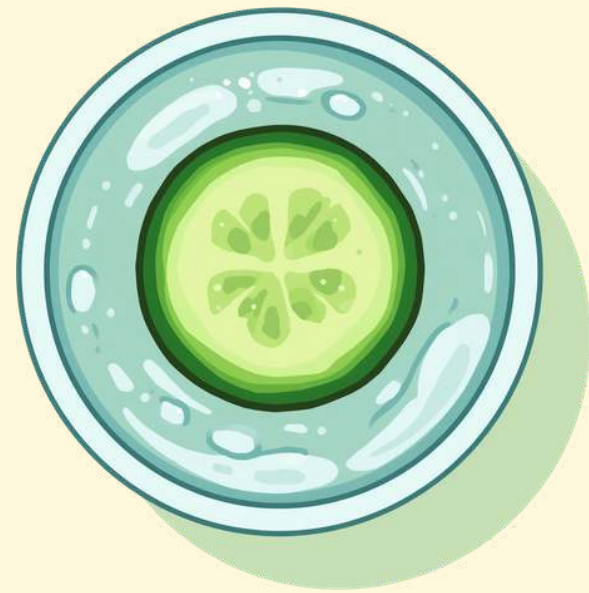


Water moves from where there is MORE water to where there
is LESS — through tiny holes in the potato called membranes.
This is Osmosis!



Osmosis All Around Us in Indian Kitchens!

Osmosis happens in many Indian foods too!



-  Cucumber slices in salt water get soft
-  Raw mango pickle (achar) shrinks
as water moves out
-  Raisins swell when soaked in water
-  Paneer dries when strained through cloth

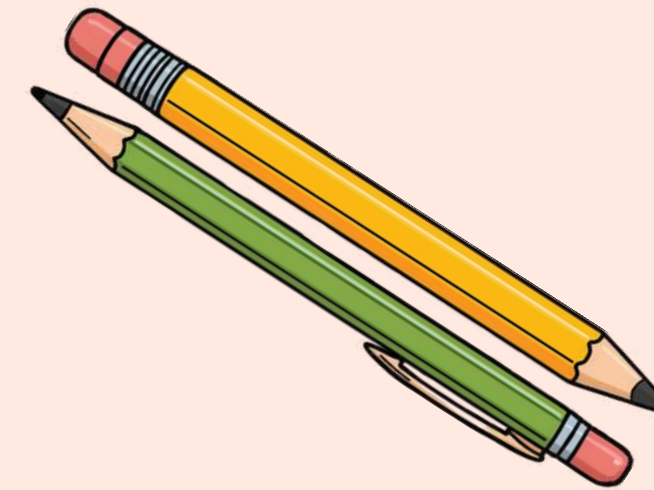




Let's Play! Quiz Time!

Drag the right answer: Why does the potato shrink in salt water?

- ☒ Water moves out of potato
- ☐ Potato eats salt
- ☐ Salt makes potato bigger
- ☐ Potato melts





Recap & Home Challenge!



Osmosis = water moves through tiny holes



Potato shrinks in salt water because water moves out



Try it at home with a potato and salt — ask your parents!



Have fun exploring osmosis!